

DATA SHEET

Three Phase Induction Motor - Squirrel Cage



Customer :

Product line : W22 NEMA Premium Efficiency Three-Phase Product code : 14874120

Frame	: 588/9T	Locked rotor time	: 41s (cold) 23s (hot)
Output	: 700 HP (515.20 kW)	Temperature rise	: 80 K
Poles	: 4	Duty cycle	: Cont.(S1)
Frequency	: 60 Hz	Ambient temperature	: -20°C to +40°C
Rated voltage	: 460 V	Altitude	: 1000 m.a.s.l.
Rated current	: 788 A	Protection degree	: IP55
L. R. Amperes	: 5122 A	Cooling method	: IC411 - TEFC
LRC	: 6.5x(Code H)	Mounting	: F-1
No load current	: 240 A	Rotation ¹	: Both (CW and CCW)
Rated speed	: 1790 rpm	Noise level ²	: 81.0 dB(A)
Slip	: 0.56 %	Starting method	: Direct On Line
Rated torque	: 2050 ft.lb	Approx. weight ³	: 5624 lb
Locked rotor torque	: 250 %		
Breakdown torque	: 220 %		
Insulation class	: F		
Service factor	: 1.00		
Moment of inertia (J)	: 335 sq.ft.lb		

Output	50%	75%	100%	Foundation loads	
Efficiency (%)	96.2	96.5	96.6	Max. traction	: 8659 lb
Power Factor	0.76	0.81	0.85	Max. compression	: 14283 lb

Losses at normative operating points (speed;torque), in percentage of rated output power

P1 (0,9;1,0)	P2 (0,5;1,0)	P3 (0,25;1,0)	P4 (0,9;0,5)	P5 (0,5;0,5)	P6 (0,5;0,25)	P7 (0,25;0,25)
4.2	3.2	2.5	2.6	1.8	1.4	0.8

	Drive end	Non drive end
Bearing type	: 6322 C3	6319 C3
Sealing	: Taconite Labyrinth	Taconite Labyrinth
Lubrication interval	: 6000 h	8000 h
Lubricant amount	: 60 g	45 g
Lubricant type	Mobil Polyrex EM	

Notes

This revision replaces and cancel the previous one, which must be eliminated.

- (1) Looking the motor from the shaft end.
- (2) Measured at 1m and with tolerance of +3dB(A).
- (3) Approximate weight subject to changes after manufacturing process.
- (4) At 100% of full load.

These are average values based on tests with sinusoidal power supply, subject to the tolerances stipulated in NEMA MG-1.

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Date	09/07/2025			

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Customer :

Space heater information

Voltage: 110-127/200-240 V

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Date	09/07/2025				

TORQUE AND CURRENT VS SPEED CURVE

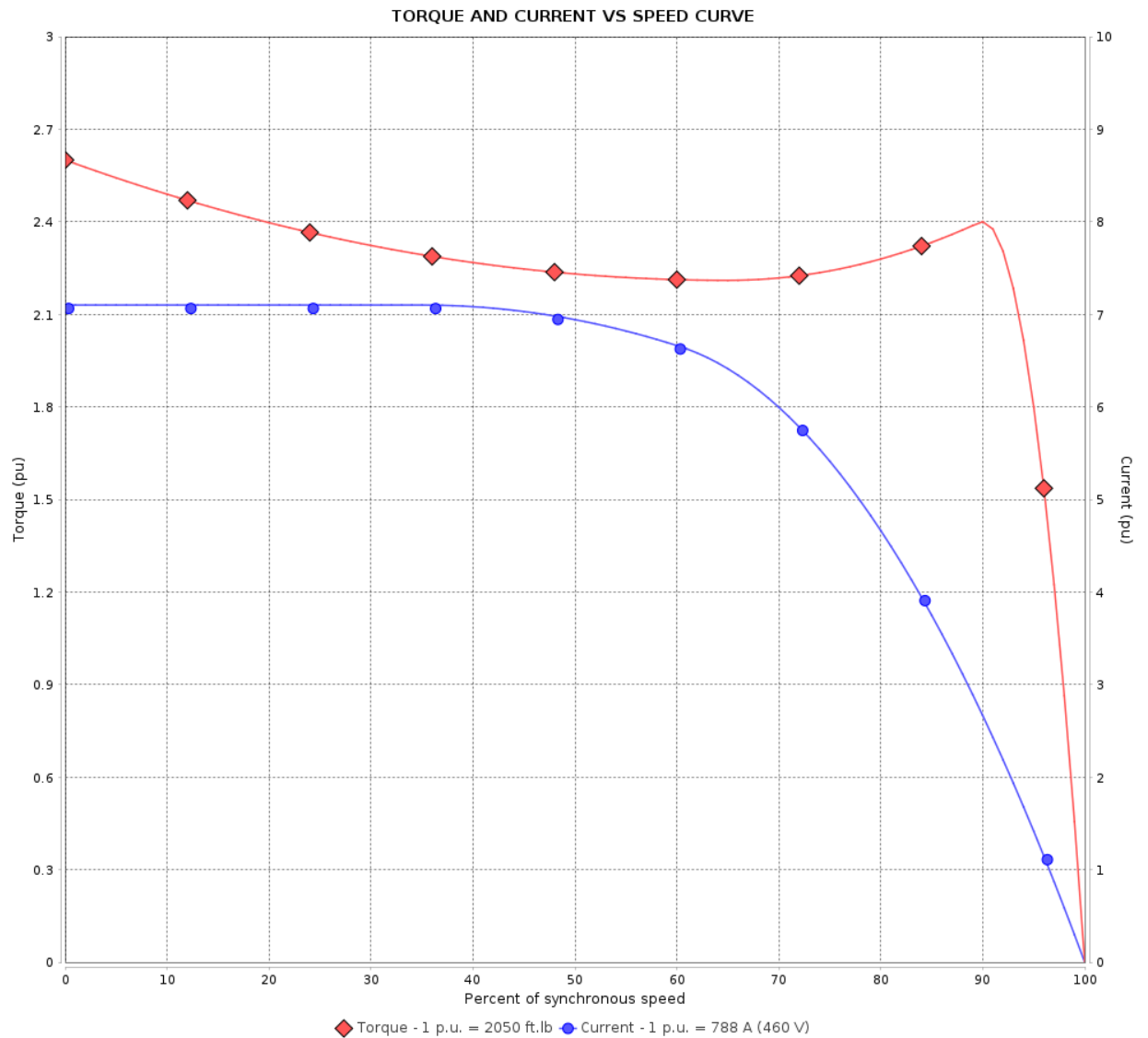
Three Phase Induction Motor - Squirrel Cage



Customer :

Product line : W22 NEMA Premium Efficiency
Three-Phase

Product code : 14874120



Performance : 460 V 60 Hz 4P

Rated current	: 788 A	Moment of inertia (J)	: 335 sq.ft.lb
LRC	: 6.5	Duty cycle	: Cont.(S1)
Rated torque	: 2050 ft.lb	Insulation class	: F
Locked rotor torque	: 250 %	Service factor	: 1.00
Breakdown torque	: 220 %	Temperature rise	: 80 K
Rated speed	: 1790 rpm		

Locked rotor time : 41s (cold) 23s (hot)

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LOAD PERFORMANCE CURVE

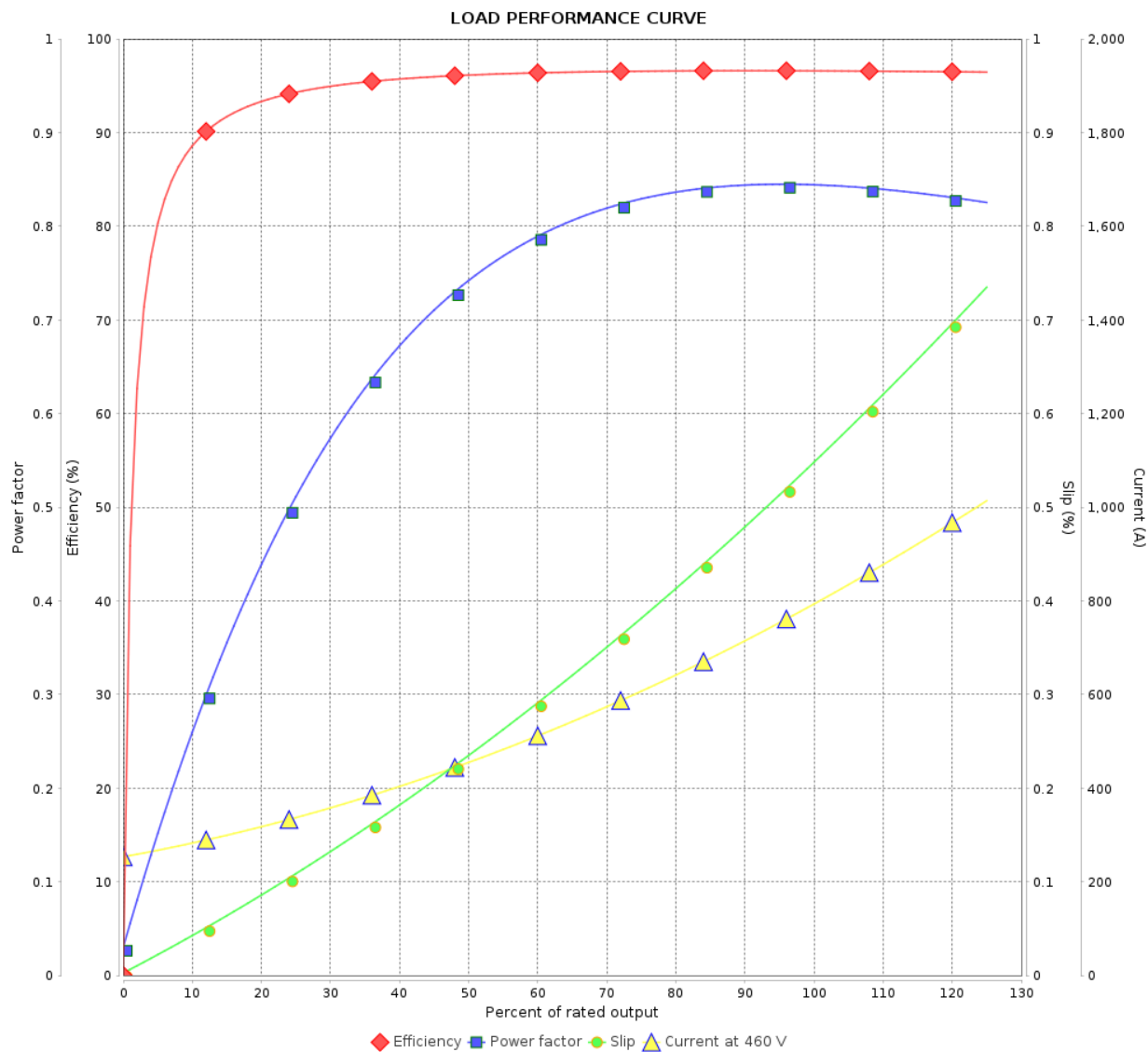
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Three-Phase

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Temperature rise : 80 K

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THERMAL LIMIT CURVE

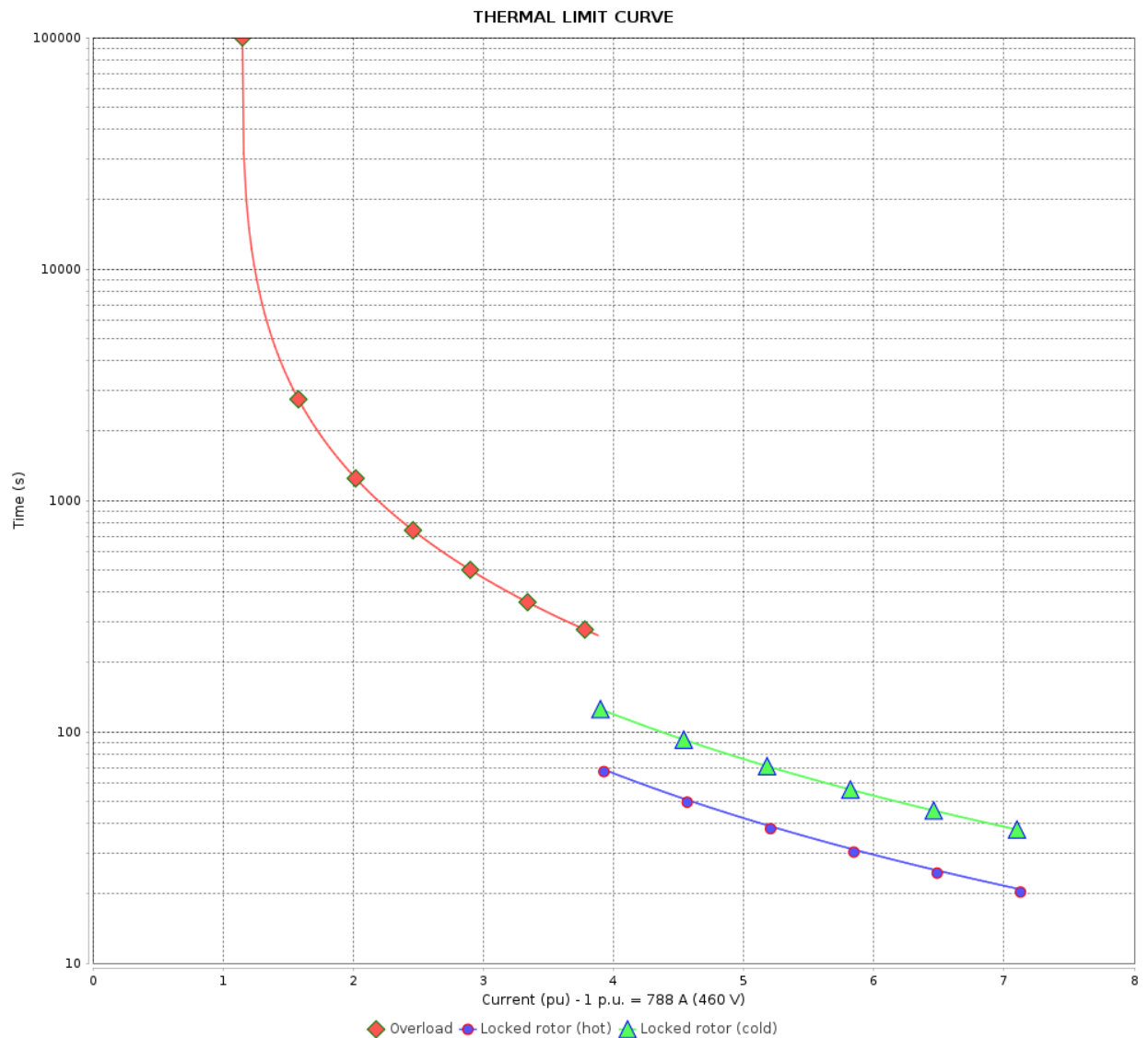
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Temperature rise : 80 K

Heating constant

Cooling constant

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VFD OPERATION CURVE

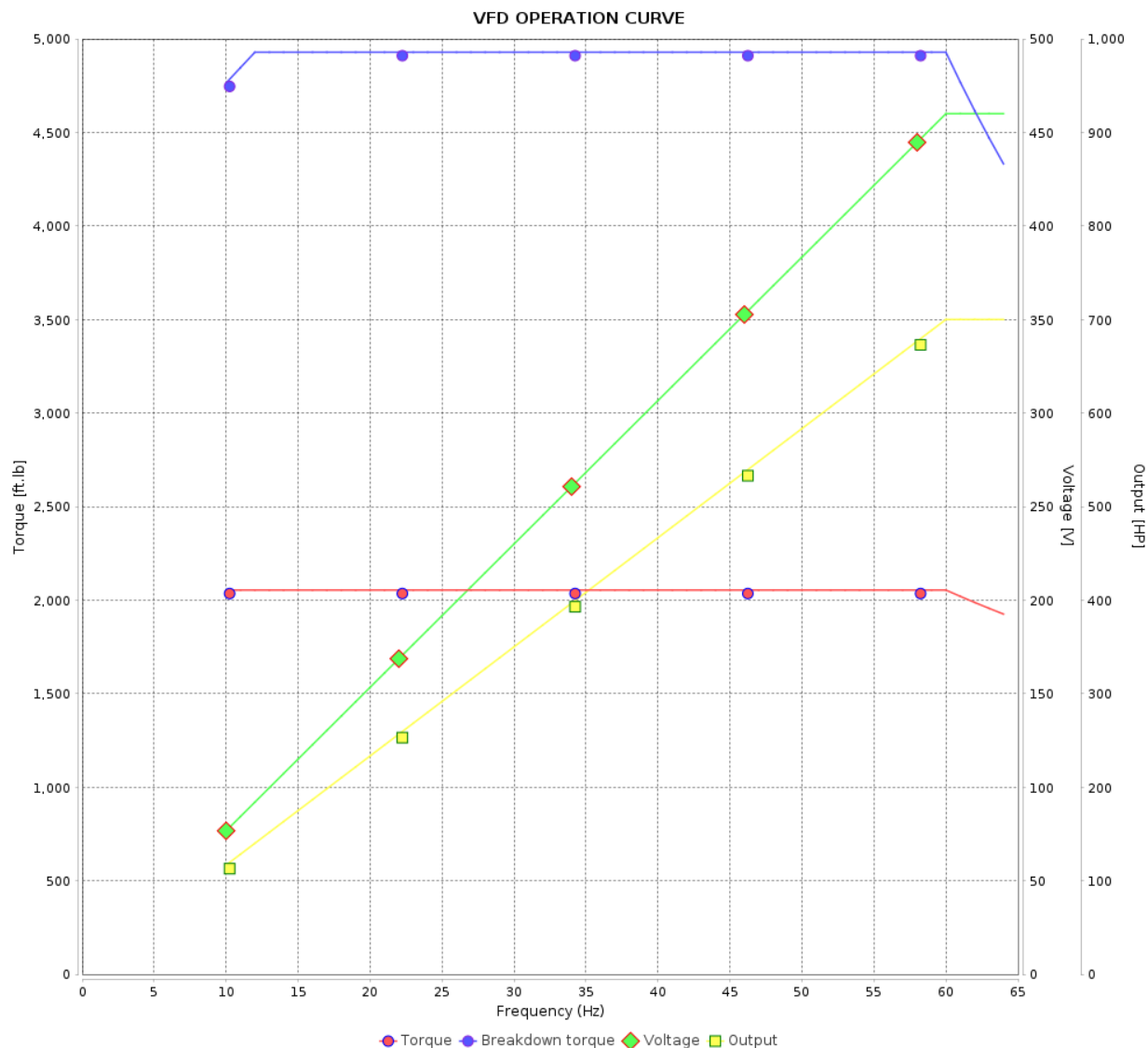
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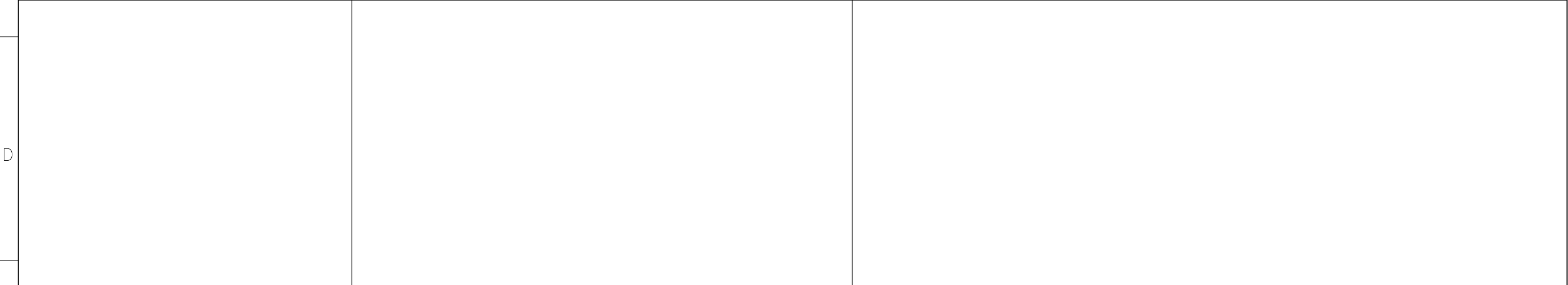
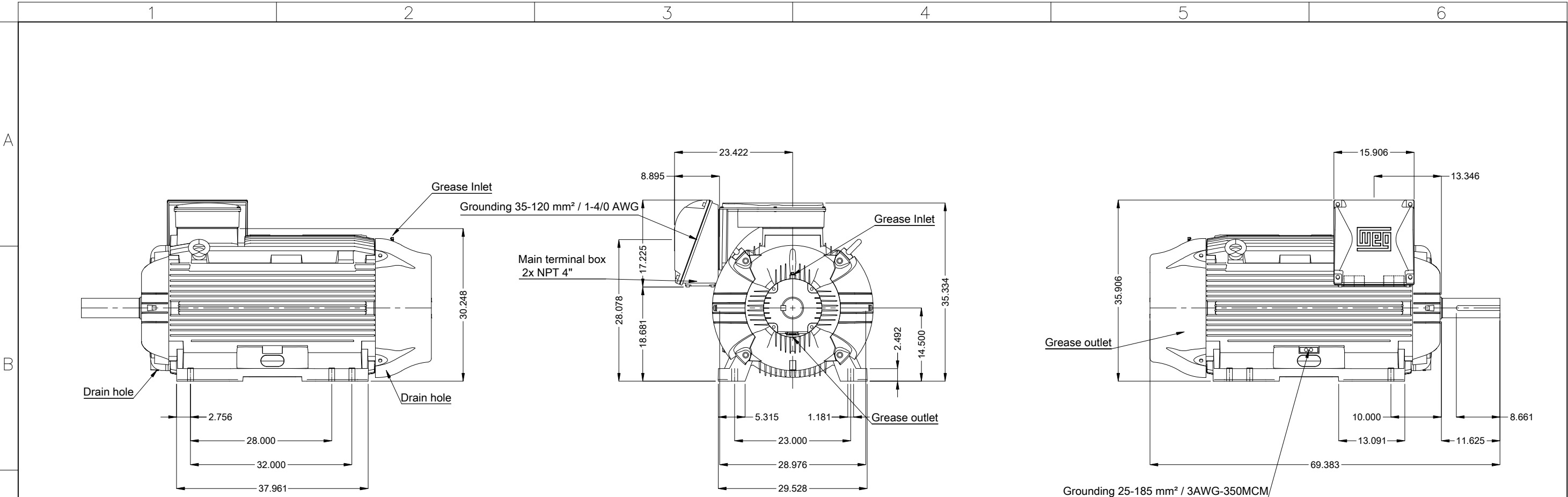


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Space heaters 110-127 V / 200-240 V		700 HP 04 Poles 60 Hz		A	
NDE endshield with electrically insulated bearing hub		Scale 1 : 20			
Drive-End Air Deflector					
Color RAL 5009					
Painting plan 203A					
Mounting NEMA F-1					
Mounting B3R(D)					
DE Shaft End		HYBRISUSER		00	
1.000 1.000 3.8750 3.8740 3.312		ECM LOC SUMMARY OF MODIFICATIONS		EXECUTED CHECKED RELEASED DATE VER	
DUNC 7/8"-9 WEG WPR-7339		EXECUTED HYBRISUSER		THREE PH. MOTOR W22 NEMA PREM. EFF.	
		CHECKED		FRAME 588/9T IP55 TEFC	
		RELEASED		PREVIEW	
		REL. DATE		WDD 00	
				SHEET 1 / 1	
				WEG	